

Customer Shopping Behavior Analysis

An End-to-End Retail Analytics Project

Python • SQL • Power BI

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1. Executive Summary

This report presents an end-to-end analysis of retail customer shopping behavior, covering 3,900 transactions across four product categories. The project combines Python for data cleaning and preparation, SQL for structured business-question analysis, and Power BI for interactive visualization, producing a set of insights intended to guide marketing, pricing, and retention strategy.

Average purchase value across the dataset is \$59.76, with male customers contributing roughly two-thirds of total revenue despite the dataset skewing toward male shoppers overall. Subscribers spend at a similar rate to non-subscribers but represent a much smaller share of the customer base, pointing to a clear growth opportunity. Loyal customers dominate the base, accounting for the vast majority of shoppers, while high discount dependency on certain product lines signals a category where margin is being traded for volume.

3,900 Transactions Analyzed	\$59.76 Avg. Purchase Amount	3.75 / 5 Avg. Review Rating
27% Customers Subscribed	80% Loyal Segment Share	18 Data Columns Analyzed

2. Project Overview

The goal of this project is to uncover actionable insights into customer spending patterns, segment behavior, product preferences, and subscription dynamics using a real-world style retail transactions dataset, and to translate those insights into concrete business recommendations.

2.1 Dataset Summary

The dataset contains 3,900 rows and 18 columns, spanning three broad categories of information:

- Customer demographics — Age, Gender, Location, Subscription Status
- Purchase details — Item Purchased, Category, Purchase Amount, Season, Size, Color
- Shopping behavior — Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

The Review Rating column contained 37 missing values, which were addressed during the cleaning stage described below.

2.2 Toolchain

- Python (pandas) — data loading, cleaning, and feature engineering
- SQL — structured analysis of ten core business questions
- Power BI — interactive dashboard for stakeholder-facing exploration

3. Data Preparation in Python

Before any business questions could be answered, the raw dataset was inspected, cleaned, and reshaped into an analysis-ready form.

3.1 Initial Structure Check

`df.info()` confirmed 3,900 non-null rows across 17 of 18 columns, with Review Rating carrying 3,863 non-null entries — the source of the 37 missing values flagged above.

```
[4]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Customer ID           3900 non-null   int64  
1   Age                   3900 non-null   int64  
2   Gender                3900 non-null   str     
3   Item Purchased        3900 non-null   str     
4   Category              3900 non-null   str     
5   Purchase Amount (USD) 3900 non-null   int64  
6   Location              3900 non-null   str     
7   Size                  3900 non-null   str     
8   Color                 3900 non-null   str     
9   Season                3900 non-null   str     
10  Review Rating          3863 non-null   float64 
11  Subscription Status    3900 non-null   str     
12  Shipping Type          3900 non-null   str     
13  Discount Applied       3900 non-null   str     
14  Promo Code Used        3900 non-null   str     
15  Previous Purchases     3900 non-null   int64  
16  Payment Method         3900 non-null   str     
17  Frequency of Purchases 3900 non-null   str     
dtypes: float64(1), int64(4), str(13)
memory usage: 835.9 KB
```

Figure 3.1 — DataFrame structure (`df.info()`)

3.2 Summary Statistics

`df.describe()` shows purchase amounts ranging from \$20 to \$100 with a mean of \$59.76 and a fairly tight standard deviation of \$23.69, indicating a broad but not extreme spread of basket sizes. Review ratings cluster tightly around a mean of 3.75, with a minimum of 2.5, suggesting no customer left an extremely negative rating.

```
[5]: df.describe()

[5]:
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

Figure 3.2 — Summary statistics (`df.describe()`)

3.3 Cleaning and Feature Engineering

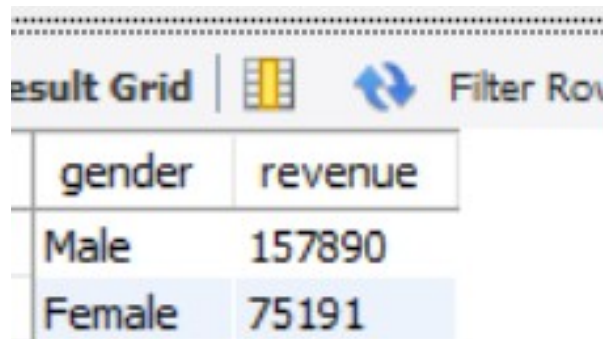
- Missing data: the 37 missing Review Rating values were imputed using the median rating for the corresponding product category.
- Column standardization: all column names were converted to snake_case for consistency and easier SQL integration.
- Feature engineering: an age_group column was created by binning customer ages, and a purchase_frequency_days column was derived from the purchase-frequency field.
- Redundancy check: Discount Applied and Promo Code Used were found to be redundant, so Promo Code Used was dropped.
- Database integration: the cleaned DataFrame was loaded into a SQL database to support the analysis in Section 4.

4. Business Analysis in SQL

With the cleaned data loaded into a SQL database, ten structured queries were written to answer specific business questions about revenue, customer segments, and product performance.

4.1 Revenue by Gender

Male customers generated \$157,890 in total revenue compared to \$75,191 from female customers — roughly 68% of total revenue coming from male shoppers, despite male transactions outnumbering female ones by a smaller margin. This points to either a larger male customer base or higher average basket size among male shoppers, and is worth validating against transaction counts.



The image shows a screenshot of a data visualization tool's 'Result Grid'. It displays a table with two columns: 'gender' and 'revenue'. The data is as follows:

gender	revenue
Male	157890
Female	75191

Figure 4.1 — Total revenue by gender

4.2 High-Spending Discount Users

839 customers used a discount while still spending above the dataset average — a meaningful segment that responds to promotions without needing them to purchase at all. These customers are strong candidates for margin-conscious, lighter-touch promotional offers rather than deep discounts.

Result Grid   Filter Rows:		
	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67
	35	91
	37	69
	40	60

Figure 4.2 — Customers who used a discount and spent above average

4.3 Top 5 Products by Rating

Gloves, Sandals, Boots, Hat, and Skirt lead the catalog by average review rating, each scoring above 3.78 out of 5. All five sit close together, suggesting rating quality is fairly consistent across the top of the catalog rather than driven by one standout product.

	item_purchased	Average Product Rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

Figure 4.3 — Top 5 products by average review rating

4.4 Shipping Type Comparison

Express shipping customers spend slightly more on average (\$60.48) than Standard shipping customers (\$58.46). The gap is modest — about \$2 — but consistent with the idea that customers willing to pay for faster delivery are marginally less price-sensitive overall.

	shipping_type	ROUND(AVG(purchase_amount),2)
▶	Express	60.48
	Standard	58.46

Figure 4.4 — Average purchase amount by shipping type

4.5 Subscribers vs. Non-Subscribers

Only 1,053 of 3,900 customers (27%) are subscribers, yet their average spend (\$59.49) is nearly identical to non-subscribers (\$59.87). Because subscribers are a minority, they contribute just \$62,645 of the \$233,081 in total revenue — well under their proportional share would suggest if subscription drove materially higher spend. This indicates subscription status is not currently a strong revenue lever, and the opportunity lies in growing subscriber count rather than assuming subscribers already spend more.

Result Grid				
		Filter Rows:	Export:	Wrap Cell Content:
	subscription_status	total_customers	avg_spend	total_revenue
►	Yes	1053	59.49	62645
	No	2847	59.87	170436

Figure 4.5 — Subscribers vs. non-subscribers: customers, average spend, and total revenue

4.6 Discount-Dependent Products

Hat, Sneakers, Coat, Sweater, and Pants show the highest discount-dependency rates, with nearly half of all purchases of these items involving a discount (47–50%). These categories carry the greatest risk of margin erosion and are good candidates for a pricing or promotion-cadence review.

	item_purchased	discount_rate
►	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

Figure 4.6 — Products with the highest discount-dependency rate

4.7 Customer Segmentation

Customers were classified into New, Returning, and Loyal segments based on purchase history. The base is heavily weighted toward Loyal customers (3,116, or 80%), with only 701 Returning and 83 New customers. This is a healthy sign of retention, but the very small New-customer pool suggests acquisition, not retention, is the more pressing growth constraint.

	customer_segment	Number of Customer
►	loyal	3116
	Returning	701
	New	83

Figure 4.7 — Customer segmentation: New, Returning, and Loyal

4.8 Top 3 Products per Category

Ranking products within each of the four categories highlights consistent leaders: Jewelry in Accessories, Blouse in Clothing, Sandals in Footwear, and Jacket in Outerwear. Order volumes across the top three items in each category are tightly clustered (typically within 10–20 orders of one another), indicating broad-based demand rather than a single hero product per category.

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

Figure 4.8 — Top 3 products by order volume, per category

4.9 Repeat Buyers and Subscriptions

Among customers with more than five previous purchases, only 958 are subscribers versus 2,518 who are not — meaning repeat buying is not strongly associated with subscribing. This reinforces the finding in Section 4.5: subscription and loyalty currently behave as largely independent dimensions, and converting high-frequency repeat buyers into subscribers is a clear, low-friction target for a retention campaign.

	subscription_status	repeat_buyer
▶	Yes	980
	No	2583

Figure 4.9 — Subscription status among repeat buyers

4.10 Revenue by Age Group

Revenue is relatively evenly spread across age groups, with Young Adults contributing the most (\$62,143) and Seniors the least (\$55,763) — a gap of roughly 11%. No single age group dominates, which suggests marketing spend can be allocated broadly rather than concentrated on one demographic, though the Young Adult segment offers the highest marginal return.

	age_group	total_revenue
▶	young_adult	62143
	middle-aged	59197
	adult	55978
	senior	55763

Figure 4.10 — Total revenue by age group

5. Interactive Dashboard in Power BI

The cleaned dataset and SQL findings were brought together into an interactive Power BI dashboard, allowing stakeholders to filter by Subscription Status, Gender, Category, and Shipping Type and explore revenue and sales patterns without needing to write SQL themselves.

The dashboard surfaces the same headline figures found in the SQL analysis — 3.9K customers, a \$59.76 average purchase amount, and a 3.75 average review rating — alongside category- and age-group-level breakdowns of revenue and sales, a subscription-status donut chart confirming the 27% / 73% split from Section 4.5, and slicer panels for every major dimension so a non-technical stakeholder can self-serve most follow-up questions without a fresh export.

6. Business Recommendations

Boost Subscriptions

Subscribers spend at essentially the same rate as non-subscribers, so the priority is growing the subscriber base rather than assuming the program itself drives higher spend. Promote exclusive, tangible benefits (early access, free express shipping) to convert high-frequency repeat buyers, who today are largely non-subscribers.

Customer Loyalty Programs

With only 701 Returning and 83 New customers against 3,116 Loyal customers, reward structures should specifically target moving Returning customers into the Loyal tier, and prioritize acquisition channels to widen the very thin New-customer pipeline.

Review Discount Policy

Hat, Sneakers, Coat, Sweater, and Pants are discounted in roughly half of all purchases. A margin review of these five lines — potentially shifting from blanket discounting to targeted offers for the high-spending discount segment identified in Section 4.2 — could recover margin without materially hurting volume.

Product Positioning

Highlight consistently top-rated items (Gloves, Sandals, Boots) and top-selling items per category (Jewelry, Blouse, Sandals, Jacket) in campaigns and homepage merchandising, since both lists show tightly clustered performance rather than one runaway winner to overexpose.

Targeted Marketing

Focus incremental marketing spend on the Young Adult segment, which generates the most revenue per age group, and on Express-shipping customers, who show a modest but consistent willingness to spend more than Standard-shipping customers.

7. Conclusion

This project demonstrates a complete analytics workflow — from raw data cleaning in Python, through structured business-question analysis in SQL, to stakeholder-facing visualization in Power BI. The resulting insights point to two consistent themes: subscription and loyalty are currently under-leveraged as revenue drivers relative to their potential, and a small set of product categories carry disproportionate discount dependency. Addressing both offers a clear, data-backed path to improving revenue and margin without requiring new data collection. All queries, cleaning steps, and the dashboard file are available in the accompanying project repository for reproducibility and further exploration.